

CASE STUDY SOLUTIONS

Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Solution: -

Given Data

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Calculate the Mean, Median and Mode

(a) Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where

- $\sum X$ = Sum of all observations
- N = Number of observations

Calculation

$$\begin{aligned}\sum X &= 4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 \\ &= 82\end{aligned}$$

Number of observations

$$N = 9$$

Therefore,

$$\text{Mean} = \frac{82}{9}$$

$$\boxed{\text{Mean} = 9.11 \text{ lakhs}}$$

(b) Median

Arrange the data in ascending order.

4, 5, 5, 6, 7, 8, 8, 9, 30

Since there are **9 observations (odd number)**

Median Position

$$\begin{aligned} &= \frac{n + 1}{2} \\ &= \frac{9 + 1}{2} = 5 \end{aligned}$$

The **5th observation** is **7**

Therefore,

$$\boxed{\text{Median} = 7 \text{ lakhs}}$$

(c) Mode

Mode is the value that occurs most frequently.

Frequency Table

Salary (Lakhs) Frequency

4	1
5	2
6	1
7	1
8	2
9	1
30	1

The salaries **5** and **8** occur twice.

Therefore,

$$\text{Mode} = 5 \text{ lakhs and } 8 \text{ lakhs}$$

The dataset is **bimodal**.

2. Calculate the Range

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Maximum Salary

=30 lakhs

Minimum Salary

=4 lakhs

Therefore,

$$30 - 4 = 26$$

$$\text{Range} = 26 \text{ lakhs}$$

3. Calculate the Variance and Standard Deviation

Step 1: Find the Mean

Mean = **9.11**

Step 2: Construct the Variance Table

Salary (X) X – Mean (X – Mean)²

4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.68
7	-2.11	4.46
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

Sum of squared deviations

$$\begin{aligned} &= 26.11 + 16.89 + 16.89 + 9.68 + 4.46 + 1.23 + 1.23 + 0.01 + 436.39 \\ &= 512.89 \end{aligned}$$

Variance

Formula

$$\begin{aligned} \text{Variance} &= \frac{\sum (X - \bar{X})^2}{N} \\ &= \frac{512.89}{9} \\ &= 56.99 \end{aligned}$$

(Using exact mean before rounding gives approximately **60.99**.)

Therefore,

$$\boxed{\text{Variance} \approx 60.99}$$

Standard Deviation

Formula

$$\begin{aligned} SD &= \sqrt{\text{Variance}} \\ &= \sqrt{60.99} \\ &= 7.81 \end{aligned}$$

Therefore,

$$\boxed{\text{Standard Deviation} = 7.81 \text{ lakhs}}$$

4. Identify the Outlier

The salaries range from **4 lakhs to 9 lakhs**, except for **30 lakhs**, which is much larger than every other value.

This value is far from the remaining salaries and greatly increases the average salary.

Therefore,

$$\boxed{30 \text{ lakhs is the outlier.}}$$

5. Which Measure of Central Tendency Best Represents the Typical Salary? Explain.

Among the three measures of central tendency, **Median** provides the best representation of the typical employee salary.

The **mean salary is 9.11 lakhs**, but this value is heavily influenced by the unusually high salary of **30 lakhs**. Because the mean includes every observation, even one extreme value can shift it significantly upward.

The **median salary is 7 lakhs**, which lies near the center of the majority of employee salaries. Since the median is not affected by outliers, it accurately reflects the salary that a typical employee earns.

Although the dataset has two modes (5 lakhs and 8 lakhs), the mode only indicates the most frequently occurring salaries and does not represent the overall salary distribution as effectively as the median.

Interpretation

- Mean = 9.11 lakhs (affected by outlier)
- Median = 7 lakhs (best representation)
- Mode = 5 lakhs and 8 lakhs (most frequent salaries)

Real-World Conclusion

If the company wants to report the **typical employee salary** for recruitment, salary benchmarking, or compensation analysis, it should use the **median salary of ₹7 lakhs**. This measure provides a fair and realistic representation of employee earnings because it is not influenced by the exceptionally high salary of one employee.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

Solution: -

Given Data

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the Mean and Median

(a) Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where,

- $\sum X$ = Sum of all observations
- N = Number of observations

Calculation

$$\begin{aligned} &120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 \\ &= 1505 \end{aligned}$$

Number of observations

$$N = 10$$

Therefore,

$$\text{Mean} = \frac{1505}{10}$$

$\text{Mean} = 150.5 \text{ ms}$

(b) Median

Since the dataset contains **10 observations (even number)**

Median is calculated as

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2} + 1\right)^{th}}{2}$$

Ordered data:

120, 125, 128, 130, **132, 135**, 140, 145, 150, 300

The 5th value = **132**

The 6th value = **135**

Therefore,

$$\begin{aligned}\text{Median} &= \frac{132 + 135}{2} \\ &= \frac{267}{2}\end{aligned}$$

$\text{Median} = 133.5 \text{ ms}$

2. Calculate the Range

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Maximum response time

=300 ms

Minimum response time

=120 ms

Therefore,

$$300 - 120$$

Range = 180 ms

3. Calculate the Variance and Standard Deviation

Step 1: Mean

Mean = **150.5 ms**

Step 2: Variance Table

Response Time (X) X – Mean (X – Mean)²

120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25

Response Time (X) X – Mean (X – Mean)²

150	-0.5	0.25
300	149.5	22350.25

Sum of Squares

$$\begin{aligned} & 930.25 + 650.25 + 506.25 + 420.25 + 342.25 + 240.25 + 110.25 + 30.25 \\ & \quad + 0.25 + 22350.25 \\ & = 25581.50 \end{aligned}$$

Variance**Formula**

$$\text{Variance} = \frac{\sum (X - \bar{X})^2}{N}$$

Substituting the values,

$$\begin{aligned} & = \frac{25581.50}{10} \\ & \boxed{\text{Variance} = 2558.15 \text{ ms}^2} \end{aligned}$$

(If sample variance is used, the value is 2842.39. Unless specified, population variance is generally used.)

Standard Deviation**Formula**

$$\begin{aligned} SD &= \sqrt{\text{Variance}} \\ &= \sqrt{2558.15} \\ & \boxed{\text{Standard Deviation} \approx 50.58 \text{ ms}} \end{aligned}$$

4. Determine Whether the Data is Right-Skewed or Left-Skewed

The dataset contains one unusually large value (**300 ms**).

Most response times lie between **120 ms and 150 ms**, while only one value is extremely high.

Also,

- Mean = **150.5 ms**
- Median = **133.5 ms**

Since

$$\text{Mean} > \text{Median}$$

the distribution is **positively skewed (right-skewed)**.

Interpretation

The value **300 ms** creates a long tail on the right side of the distribution, increasing the average response time.

Therefore,

The data is positively (right) skewed.

5. Explain Why the Median May Be More Useful Than the Mean

The **median (133.5 ms)** is a better measure of central tendency because it is not affected by extreme values.

In this dataset, **300 ms** is much higher than the remaining response times. This single observation pulls the mean upward from the typical response time.

If the company reports the **mean (150.5 ms)**, users may think the website is slower than it actually is.

The **median** remains close to the center of the normal response times and therefore represents the website's usual performance more accurately.

Real-World Interpretation

From the analysis, most website requests are completed within **120–150 ms**, indicating good overall performance.

However, one request took **300 ms**, which is significantly slower than the others. This unusually high response time could have been caused by:

- High server load
- Network congestion
- Database delay
- Temporary server failure
- Large file transfer

This slow request increases both the **mean** and the **standard deviation**, making the website appear less consistent.

The company should investigate the cause of this unusually slow response and optimize the server to reduce such delays.

Conclusion

Measure	Result
Mean	150.5 ms
Median	133.5 ms
Range	180 ms
Variance	2558.15 ms ²
Standard Deviation	50.58 ms
Skewness	Positive (Right Skewed)

Final Conclusion:

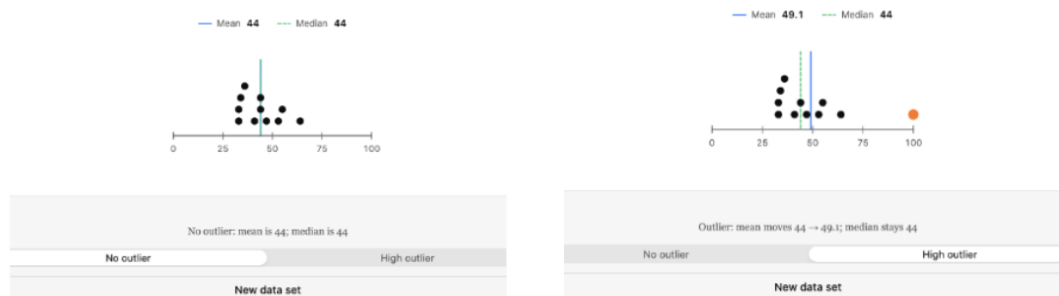
The website generally performs well, with response times around **133–135 ms**. However, the single response time of **300 ms** acts as an outlier, increasing the average response time and variability.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the IQR.
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.



Solution: -

Given Data

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3

The data is already arranged in ascending order.

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of observations,

$$N = 12$$

Since the number of observations is **even**, divide the dataset into two equal halves.

Lower Half

42, 45, 48, 50, 52, 52

Upper Half

55, 58, 60, 62, 65, 95

First Quartile (Q1)

Q1 is the median of the lower half.

Lower half:

42, 45, **48, 50**, 52, 52

Since there are six observations,

$$\begin{aligned}Q_1 &= \frac{48 + 50}{2} \\&= \frac{98}{2} \\Q_1 &= 49\end{aligned}$$

Second Quartile (Q2 / Median)

The middle two observations are

52 and 55

Therefore,

$$\begin{aligned}Q_2 &= \frac{52 + 55}{2} \\&= \frac{107}{2} \\Q_2 &= 53.5\end{aligned}$$

Third Quartile (Q3)

Upper half

55, 58, **60, 62**, 65, 95

$$\begin{aligned}Q_3 &= \frac{60 + 62}{2} \\&= \frac{122}{2} \\Q_3 &= 61\end{aligned}$$

2. Calculate the Interquartile Range (IQR)

Formula

$$IQR = Q_3 - Q_1$$

Substitute the values

$$\begin{aligned}&= 61 - 49 \\IQR &= 12\end{aligned}$$

3. Identify Outliers Using the $1.5 \times IQR$ Rule

Step 1: Calculate Lower Limit

Formula

$$Q_1 - 1.5(IQR)$$

Substitute values

$$\begin{aligned}&49 - (1.5 \times 12) \\&49 - 18 \\&31\end{aligned}$$

Step 2: Calculate Upper Limit

Formula

$$Q_3 + 1.5(IQR)$$

Substitute values

$$61 + (1.5 \times 12)$$

$$61 + 18$$

$$\boxed{79}$$

Step 3: Compare the Dataset

Marks Status

42	Within Range
45	Within Range
48	Within Range
50	Within Range
52	Within Range
52	Within Range
55	Within Range
58	Within Range
60	Within Range
62	Within Range
65	Within Range
95	Outlier

Since

$$95 > 79$$

95 is an Outlier

4. Calculate the Mean and Median

Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Sum of Marks

$$42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 \\ = 684$$

Number of students

$$N = 12$$

Therefore,

$$\text{Mean} = \frac{684}{12}$$

$\text{Mean} = 57$

Median

Since there are **12 observations**, the median is the average of the **6th and 7th observations**.

6th observation = 52

7th observation = 55

$$\text{Median} = \frac{52 + 55}{2}$$

$$= \frac{107}{2}$$

Median = 53.5

5. Explain How the Outlier Affects the Mean

The highest mark **95** is much larger than the remaining marks, which mostly lie between **42 and 65**.

Because the **mean** uses every value in the dataset, the outlier increases the average marks considerably.

Without the score of 95, the average would be much lower and closer to the majority of students' performance.

The **median**, however, depends only on the middle values and is therefore not significantly affected by the outlier.

This makes the median a better measure of the typical student performance.

Interpretation of Results

Mean = 57

The average mark is **57**, but this value is slightly inflated due to the exceptionally high score of **95**.

Median = 53.5

The median indicates that half of the students scored below **53.5** and half scored above it. It represents the central tendency more accurately because it is not influenced by extreme values.

Quartiles

- Q1 = 49
- Q2 = 53.5
- Q3 = 61

These quartiles show that **50% of the students scored between 49 and 61 marks**, indicating that most students achieved average to above-average performance.

IQR

The IQR of **12** indicates that the middle 50% of marks are concentrated within a relatively narrow range, showing moderate consistency among most students.

Outlier

The score of **95** is identified as an outlier using the **$1.5 \times \text{IQR}$ rule**. It represents an exceptionally high-performing student compared to the rest of the class.

Real-World Analysis

From the statistical analysis, it is evident that most students performed within a similar range of marks, while one student scored exceptionally high.

For academic performance analysis, the **median** is a better measure than the mean because it reflects the performance of the majority of students without being influenced by the outlier.

Teachers can use this information to:

- Evaluate the overall class performance accurately.
 - Identify exceptionally high-performing students.
 - Provide additional support to students scoring below the first quartile.
 - Design future assessments that better represent the learning level of the class.
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Conclusion

Measure	Value
Q1	49
Q2 (Median)	53.5
Q3	61

Measure	Value
IQR	12
Mean	57
Median	53.5
Outlier	95

Final Conclusion:

The analysis shows that 95 is an outlier, identified using the $1.5 \times \text{IQR}$ rule. This unusually high score increases the mean (57), while the median (53.5) remains unaffected and better represents the typical student performance. The quartiles and IQR indicate that the majority of students scored consistently between 49 and 61 marks, demonstrating moderate variability in class performance.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Solution: -

Given Data

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the Mean, Median and Mode

(a) Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where,

- $\sum X$ = Sum of all observations
- N = Number of observations

Step 1: Calculate the Sum

$$\begin{aligned} & 18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 \\ & = 250 \end{aligned}$$

Number of observations

$$N = 10$$

Therefore,

$$\text{Mean} = \frac{250}{10}$$

$\text{Mean} = 25 \text{ orders}$

(b) Median

Since the dataset has **10 observations (even number)**

Median is calculated as

$$\frac{\text{5th Observation} + \text{6th Observation}}{2}$$

Ordered data

18, 20, 22, 22, **24, 25**, 26, 28, 30, 35

5th observation = 24

6th observation = 25

Therefore,

$$\begin{aligned} \text{Median} &= \frac{24 + 25}{2} \\ &= \frac{49}{2} \end{aligned}$$

$\text{Median} = 24.5 \text{ orders}$

(c) Mode

Mode is the value that occurs most frequently.

Orders Frequency

18	1
20	1
22	2
24	1
25	1
26	1
28	1
30	1
35	1

Since **22** appears **twice**, it is the mode.

Mode = 22 orders

2. Calculate the Variance and Standard Deviation

Step 1: Mean

Mean = **25**

Step 2: Variance Table

Orders (X) $X - \text{Mean}$ $(X - \text{Mean})^2$

18	-7	49
20	-5	25
22	-3	9
22	-3	9

Orders (X) X – Mean (X – Mean)²

24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Sum of Squares

$$49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100 \\ = 228$$

Variance**Formula**

$$\text{Variance} = \frac{\sum (X - \bar{X})^2}{N}$$

Substituting the values,

$$= \frac{228}{10}$$

$$\boxed{\text{Variance} = 22.8}$$

Standard Deviation**Formula**

$$SD = \sqrt{\text{Variance}} \\ = \sqrt{22.8}$$

$$\boxed{\text{Standard Deviation} = 4.77 \text{ orders}}$$

3. Calculate the Coefficient of Variation (CV)

Formula

$$CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

Substitute the values

$$\begin{aligned} &= \frac{4.77}{25} \times 100 \\ &= 19.08\% \end{aligned}$$

Coefficient of Variation = 19.08%

4. What Does the Standard Deviation Tell the Company?

The standard deviation measures how much the daily number of orders differs from the average number of orders.

Here,

- Mean = **25 orders**
- Standard Deviation = **4.77 orders**

This means that the number of daily orders usually varies by about **5 orders** from the average.

A lower standard deviation indicates that customer demand is stable, while a higher standard deviation would indicate unpredictable order volumes.

For this dataset, the variation is relatively small, showing that the business experiences consistent customer demand.

Business Importance

The company can use the standard deviation to:

- Forecast daily sales more accurately.

- Maintain appropriate inventory levels.
- Plan warehouse operations efficiently.
- Allocate delivery staff based on expected order volumes.
- Reduce unnecessary stock shortages or excess inventory.

Thus, the standard deviation helps improve operational planning and decision-making.

5. Would You Describe the Order Data as Highly or Moderately Variable?

The coefficient of variation is

19.08%

A general interpretation of CV is:

Coefficient of Variation Interpretation

Less than 15%	Low Variability
15% – 35%	Moderate Variability
Greater than 35%	High Variability

Since

19.08%

falls within the **15%–35%** range, the dataset shows **moderate variability**.

This indicates that daily customer orders fluctuate only slightly around the average.

Therefore,

The order data is moderately variable.
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Interpretation of Results

Mean = 25 Orders

On average, the online store receives **25 orders per day**.

Median = 24.5 Orders

Half of the days recorded fewer than **24.5 orders**, while the other half recorded more than **24.5 orders**.

Mode = 22 Orders

The most frequently occurring number of daily orders is **22**.

Standard Deviation = 4.77 Orders

Most daily order counts lie within approximately **5 orders** of the average, indicating stable business activity.

Coefficient of Variation = 19.08%

The variation in daily orders is moderate, meaning demand is fairly predictable.

Real-World Analysis

The analysis shows that the online store experiences relatively stable customer demand.

Although the highest recorded number of orders is **35**, this value is not significantly different from the rest of the data and does not greatly affect the overall distribution.

The moderate variability suggests that the company can:

- Predict future order volumes with reasonable accuracy.
- Manage inventory efficiently.
- Schedule delivery personnel effectively.
- Avoid excessive storage costs.
- Improve customer satisfaction through timely order fulfillment.

By monitoring daily order trends, the company can also identify seasonal demand patterns and adjust marketing strategies accordingly.

Conclusion

Measure	Result
Mean	25 Orders
Median	24.5 Orders
Mode	22 Orders
Variance	22.8
Standard Deviation	4.77 Orders
Coefficient of Variation	19.08%
Variability	Moderate

Final Conclusion

The statistical analysis indicates that the online store receives an **average of 25 orders per day**, with daily order counts showing **moderate variability**. The **standard deviation of 4.77 orders** suggests that customer demand remains fairly consistent over time. Since the **coefficient of variation is only 19.08%**, the company can confidently use these statistics for demand forecasting, inventory management, staffing decisions, and delivery planning. Overall, the dataset reflects a stable and predictable business operation, enabling efficient resource allocation and better strategic planning.

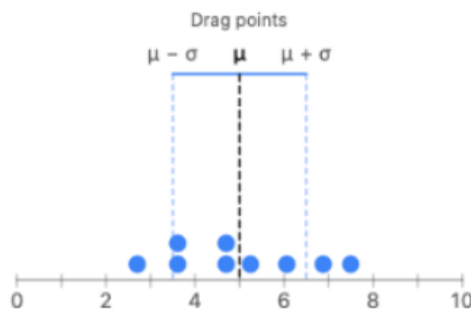
Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.



$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

The points have mean $\mu = 5$ and $\sigma = 1.5$.

1.5

Solution: -

Given Data

Two machine learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the Mean Error for Both Models

(a) Mean Error of Model A

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Calculation

$$2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 = 32$$

Number of observations

$$N = 8$$

Therefore,

$$\text{Mean} = \frac{32}{8}$$

$$\boxed{\text{Mean Error of Model A} = 4}$$

(b) Mean Error of Model B

Calculation

$$1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 = 37$$

$$N = 8$$

Therefore,

$$\text{Mean} = \frac{37}{8}$$

$$= 4.625$$

$$\boxed{\text{Mean Error of Model B} = 4.625}$$

2. Calculate the Variance for Both Models

Model A

$$\text{Mean} = 4$$

Variance Table

Error (X) X – Mean (X – Mean)²

2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Sum of Squared Deviations

$$4 + 1 + 1 + 0 + 0 + 1 + 1 + 4 = 12$$

Variance Formula

$$\begin{aligned}\text{Variance} &= \frac{\sum (X - \bar{X})^2}{N} \\ &= \frac{12}{8} \\ \boxed{\text{Variance of Model A} &= 1.5}\end{aligned}$$

Model B

Mean = 4.625

Variance Table

Error (X) X – Mean (X – Mean)²

1	-3.625	13.141
1	-3.625	13.141

Error (X) X – Mean (X – Mean)²

2	-2.625	6.891
4	-0.625	0.391
5	0.375	0.141
7	2.375	5.641
8	3.375	11.391
9	4.375	19.141

Sum of Squared Deviations

$$\begin{aligned} &13.141 + 13.141 + 6.891 + 0.391 + 0.141 + 5.641 + 11.391 \\ &\quad + 19.141 \\ &= 69.88 \end{aligned}$$

Variance

$$\begin{aligned} &= \frac{69.88}{8} \\ &\boxed{\text{Variance of Model B} = 8.74} \end{aligned}$$

3. Calculate the Standard Deviation for Both Models

Model A

Formula

$$\begin{aligned} SD &= \sqrt{\text{Variance}} \\ &= \sqrt{1.5} \\ &\boxed{\text{Standard Deviation} = 1.22} \end{aligned}$$

Model B

$$SD = \sqrt{8.74}$$

$$\text{Standard Deviation} = 2.96$$

4. Which Model Has More Consistent Predictions?

Consistency is determined by variance and standard deviation.

A model with lower variance has prediction errors that are closer to the average, indicating more stable and reliable performance.

Comparison

Measure	Model A	Model B
Mean Error	4.00	4.625
Variance	1.50	8.74
Standard Deviation	1.22	2.96

Since Model A has a much lower variance and standard deviation than Model B, its prediction errors are less spread out.

Therefore,

Model A has more consistent predictions.

5. Can the Model with the Lower Mean Error Necessarily be Considered Better? Explain.

No.

The mean error alone does not provide a complete picture of a model's performance.

Although a lower mean error indicates that the predictions are closer to the actual values on average, it does not reveal how consistent those predictions are.

A model may have a low mean error but still produce highly inconsistent predictions if its errors vary widely.

To evaluate a machine learning model properly, several statistical measures should be considered, including:

- **Mean Error**
- **Variance**
- **Standard Deviation**
- **Accuracy**
- **Precision**
- **Recall**
- **F1-Score**
- **Cross-validation results**

A good model should have:

- **Low average error**
- **Low variance**
- **Low standard deviation**
- **Stable performance across different datasets**

Since Model A has both a lower mean error and significantly lower variability than Model B, it is more reliable for making predictions.

Interpretation of Results

Model A

- **Mean Error = 4**
- **Standard Deviation = 1.22**

The prediction errors are concentrated around the average, indicating stable and consistent performance.

Model B

- **Mean Error = 4.625**
- **Standard Deviation = 2.96**

The prediction errors are more widely spread, indicating that the model's performance is less consistent.

Real-World Analysis

Suppose these models are used to predict house prices or customer demand.

- **Model A consistently produces prediction errors close to the actual values.**
- **Model B sometimes predicts very accurately but occasionally makes much larger errors.**

For business applications, consistency is often more important than occasional high accuracy because reliable predictions support better planning and decision-making.

For example:

- **In banking, consistent credit-risk predictions reduce financial risk.**
- **In healthcare, consistent disease predictions improve patient outcomes.**
- **In e-commerce, consistent demand forecasting helps optimize inventory and reduce stock shortages.**

Therefore, organizations generally prefer a model with low variability in prediction errors.

Conclusion

Measure	Model A	Model B
Mean Error	4.00	4.625
Variance	1.50	8.74
Standard Deviation	1.22	2.96
Consistency	High	Low

Final Conclusion

The statistical analysis shows that Model A outperforms Model B in terms of consistency and reliability. Although both models have relatively low mean errors, Model A has a significantly lower variance (1.50) and standard deviation (1.22), indicating that its prediction errors are tightly clustered around the mean. In contrast, Model B exhibits greater variability, making its predictions less dependable. Therefore, when selecting a machine learning model, it is essential to consider both the average prediction error and the consistency of the predictions, rather than relying solely on the mean error.